

Exhaustive verification of Seymour’s Second Neighborhood Conjecture for oriented graphs on at most 16 vertices*

Automated verification pipeline (Atris workspace)

July 23, 2026

Abstract

Seymour’s Second Neighborhood Conjecture states that every oriented graph (finite simple digraph with no loops and no 2-cycles) contains a vertex v with $|N^{++}(v)| \geq |N^+(v)|$, where $N^{++}(v)$ is the set of vertices at directed distance exactly two from v . We report an exhaustive computational verification, fully machine-verified for all oriented graphs on at most 16 vertices: no counterexample exists on $n \leq 16$ vertices. The result rests on 47 unsatisfiable SAT instances solved by CaDiCaL 3.0.1 with DRAT proof logging, every proof checked by drat-trim; the instances cover the search space through a minimum out-degree sharding scheme whose covering and symmetry-breaking arguments are short pen-and-paper proofs reproduced here. (The full artifact set spans $n \leq 18$ and 63 CNF instances; all 59 completed refutations are drat-trim-verified, including the two largest proofs, 0.60 GB checked in 1,689s and 4.23 GB checked in 9,671s; the 47 form the $n \leq 16$ claim.) Independently of the SAT chain, all oriented graphs on $n \leq 8$ vertices (575,016,219 at $n = 8$) were enumerated with nauty and checked by a from-scratch C program, with per-order counts matching OEIS A001174 exactly and zero counterexamples. At $n = 17, 18$ the six lowest-degree shards are likewise refuted and verified; the near-tournament top shards hit the solver cap with no verdict and remain open as machine results, except that the $n = 17$ tournament shard is closed by Fisher’s theorem, so those orders are reported separately and not claimed. We found no previously published exhaustive verification of the conjecture over all oriented graphs of a given order, so we state the small-order record here explicitly rather than compare against one.

1 The conjecture and its history

Let D be an *oriented graph*: a finite directed graph with no loops, no parallel arcs, and no digons (if (u, v) is an arc then (v, u) is not). For a vertex v write $N^+(v)$ for its out-neighborhood and $N^{++}(v)$ for its *second out-neighborhood*, the set of vertices at directed distance exactly 2 from v . A vertex v with $|N^{++}(v)| \geq |N^+(v)|$ is called a *Seymour vertex*.

Conjecture 1 (Seymour, c. 1990; see Dean and Latka [1]). *Every oriented graph has a Seymour vertex.*

*Working note, not peer reviewed. The computations and this text were produced by an automated agent pipeline (Claude Fable 5, Atris workspace of Keshav Rao) on a single shared Apple Silicon laptop, 2026-07-22/23. Section 6 lists exactly what a human reviewer must audit before any of this is submitted anywhere. No claim in this note should be trusted beyond the evidence described in it.

A counterexample is an oriented graph in which *every* vertex v satisfies $|N^{++}(v)| < |N^+(v)|$. The conjecture is open. The main published partial results are:

- **Tournaments.** Dean’s conjecture (the tournament case) was proved by Fisher [2] using an averaging argument over probability distributions; Havet and Thomassé [3] gave a combinatorial proof via median orders, producing two Seymour vertices when the tournament has no dominated vertex.
- **Small minimum out-degree.** Kaneko and Locke [4] proved the conjecture for oriented graphs with minimum out-degree at most 6. A June 2026 preprint by Sadhukhan, Sandeep, and Sen [10] claims the case of minimum out-degree exactly 7, using local reductions plus computer-assisted infeasibility checks; it has not, to our knowledge, been refereed.
- **Approximate versions.** Chen, Shen, and Yuster [5] proved every oriented graph has a vertex with $|N^{++}(v)| \geq \gamma |N^+(v)|$ for $\gamma = 0.657\dots$, with later improvements to the constant by others.
- **Random and dense settings.** Cohn, Godbole, Wright Harkness, and Zhang [8] showed the conjecture holds asymptotically almost surely for random tournaments and random digraphs. Espuny Díaz, Girão, Granet, and Kronenberg [9] proved that a.a.s. every orientation of $G(n, p)$ satisfies the conjecture for fixed $p < 1/2$, and gave two reductions; their Proposition 4 shows a vertex-minimal counterexample D must have minimum out-degree greater than $\sqrt{|V(D)|}$.
- **Structural classes.** Fidler and Yuster [6] handled tournaments minus a star or a subtournament and digraphs of minimum degree $|V(D)| - 2$; Brantner, Brockman, Kay, and Snively [7] handled digraphs whose underlying graph is triangle-free.

On the exhaustive small-order record. We searched for a published exhaustive verification of Conjecture 1 over *all* oriented graphs on $n \leq k$ vertices for any k and found none: neither the standard open-problem surveys [11, 12] nor the papers above report one. An earlier internal working note of ours asserted a published record of $n \leq 7$; we could not trace that assertion to any source and we withdraw it. The correct statement appears to be that no such record was published, presumably because checks up to $n \approx 8$ are routine with nauty. The contribution of this note is therefore not the beating of a record but the explicit, certificate-backed statement of one.

Theorem 1 (main claim, machine-checked). *No oriented graph on at most 16 vertices is a counterexample to Conjecture 1.*

Kaneko and Locke [4] (any counterexample has minimum out-degree at least 7) together with the arc-counting bound of Section 2.2 (minimum out-degree at most $\lfloor (n-1)/2 \rfloor$) already force $n \geq 15$ for a hypothetical counterexample; combined with Theorem 1 it must have $n \geq 17$. Theorem 1 itself is the unconditional statement that does not rely on [4]: our shards re-prove the low-degree cases for $n \leq 16$ from scratch rather than citing them.

2 Method

The overall shape: for each order n , we encode “an oriented graph on n labeled vertices in which every vertex violates the Seymour inequality” as a CNF formula, split the space into minimum

out-degree shards with symmetry breaking, obtain UNSAT verdicts with DRAT proofs from CaDiCaL 3.0.1 [13], and check every proof with drat-trim [14]. The DRAT certificates prove the CNFs unsatisfiable; the link from the CNFs to the conjecture is the soundness chain below, which is deliberately short and must be audited by a human.

2.1 Encoding

The encoder is `encode.py` (about 150 lines, PySAT [15] 1.9). Variables: x_{uv} for each ordered pair $(u \rightarrow v \text{ arc})$, and y_{vw} for each ordered pair with $v \neq w$, an *upper indicator* for $w \in N^{++}(v)$. Clauses:

1. $(\neg x_{uv} \vee \neg x_{vu})$ for all $u < v$: no digons. No variable x_{vv} exists, so loops cannot be represented.
2. $(x_{vu} \wedge x_{uw} \wedge \neg x_{vw}) \rightarrow y_{vw}$ for all distinct u, v, w : every genuine 2-path not shortcut by an arc forces y_{vw} up. Hence in every model, $\sum_w y_{vw} \geq |N^{++}(v)|$.
3. For each v : a totalizer cardinality constraint [16] encoding $\sum_w y_{vw} + \sum_u \neg x_{vu} \leq n - 2$, which given the previous point is equivalent to $|N^{++}(v)| \leq |N^+(v)| - 1$, i.e., v violates the Seymour inequality.

Completeness (the direction that matters for a nonexistence claim): if a counterexample D on n labeled vertices exists, set x from the arcs of D and set y_{vw} to the exact indicator of $w \in N^{++}(v)$. Group (1) holds because D is an oriented graph; group (2) holds because any 2-path $v \rightarrow u \rightarrow w$ with no arc $v \rightarrow w$ witnesses $w \in N^{++}(v)$; group (3) holds because every vertex of D violates the inequality. So the CNF is satisfiable, and UNSAT verdicts rule out every labeled counterexample on n vertices. Spurious $y = 1$ assignments only tighten constraint (3), so soundness of the search is not an issue for the nonexistence direction.

For $n \leq 9$ we ran this formula over the whole space with no symmetry breaking and a single additional constraint, minimum out-degree ≥ 1 (the encoder's default): this excludes only graphs containing a sink, and a sink v has $|N^{++}(v)| = 0 = |N^+(v)|$, so it is itself a Seymour vertex and no graph containing one is a counterexample. Beyond that one elementary observation the four instances are assumption-free: their verified refutations alone prove Theorem 1 for $n \leq 9$.

2.2 Degree shards and symmetry breaking

For $n \geq 10$ the whole-space formula becomes slow, so the space is split. In a counterexample no vertex is a sink (a sink v has $|N^{++}(v)| = 0 = |N^+(v)|$ and would be a Seymour vertex), so the minimum out-degree d_0 satisfies $d_0 \geq 1$; counting arcs, $n d_0 \leq \sum_v \deg^+(v) = |A(D)| \leq \binom{n}{2}$, so $d_0 \leq \lfloor (n-1)/2 \rfloor$. Shard s_d (for $1 \leq d \leq \lfloor (n-1)/2 \rfloor$) asserts:

- every vertex has out-degree $\geq d$ (totalizers),
- vertex 0 has out-degree exactly d (totalizers),
- $N^+(0) = \{1, \dots, d\}$ (unit clauses),
- two-block lex-leader symmetry breaking: for every adjacent transposition σ acting within $\{1, \dots, d\}$ or within $\{d+1, \dots, n-1\}$, the adjacency matrix A satisfies $A \leq_{\text{lex}} A \circ \sigma$ [17].

Covering argument. Suppose a counterexample D on n vertices exists and let d_0 be its minimum out-degree, so $1 \leq d_0 \leq \lfloor (n-1)/2 \rfloor$. Relabel a minimum out-degree vertex to 0 and its out-neighbors to $1, \dots, d_0$. The residual relabeling freedom is exactly the subgroup $H = \text{Sym}\{1, \dots, d_0\} \times \text{Sym}\{d_0 + 1, \dots, n-1\}$, which preserves the pinned constraints. Choose the labeling whose adjacency matrix is lexicographically minimal over the H -orbit; it satisfies $A \leq_{\text{lex}} A \circ \sigma$ for every $\sigma \in H$, in particular for the encoded adjacent transpositions. Hence some assignment satisfies shard s_{d_0} , and UNSAT verdicts on all shards of order n rule out all counterexamples on n vertices. Note the encoded constraints are a *subset* (adjacent transpositions only) of the full lex-leader condition; using a subset weakens the formula and therefore cannot cause a false UNSAT.

For $n = 10$ we additionally ran the whole space in one piece (minimum out-degree ≥ 1 , full-lex symmetry breaking over all adjacent transpositions of the symmetric group, which is sound because the formula without pinning is invariant under all vertex permutations), so Theorem 1 at $n = 10$ does not depend on the shard argument either. Cross-checks at $n = 10, 11$ with the (audited, elementary) bound “any counterexample has minimum out-degree ≥ 3 ” agree; that lemma is stated in the artifact repository but is not load-bearing anywhere in this note, precisely because shards s_1 and s_2 were machine-proved directly.

2.3 Tournament-forced shards and Lemma T

For odd n , the top shard $d = (n-1)/2$ forces $\sum_v \deg^+(v) \geq n(n-1)/2$, i.e., every pair adjacent: the graph is a $\frac{n-1}{2}$ -regular tournament. These shards are closed twice over:

1. By Fisher’s theorem [2]: every tournament has a Seymour vertex, so no tournament is a counterexample. This is published, peer-reviewed mathematics and is the citation we prefer.
2. By the following elementary argument, produced in-session and **not yet reviewed by any human**; we record it because it is short enough to audit in one sitting.

Lemma 1 (Lemma T, **unreviewed**). *In a k -regular tournament, every vertex v satisfies $|N^{++}(v)| = |N^+(v)| = k$, so no such tournament is a counterexample.*

Proof (to be audited). Fix v and take any $w \notin \{v\} \cup N^+(v)$. If $w \notin N^{++}(v)$, then no $u \in N^+(v)$ has $u \rightarrow w$; by totality of the tournament, $w \rightarrow u$ for all $u \in N^+(v)$, and also $w \rightarrow v$ (since $v \rightarrow w$ would put w in $N^+(v)$). Then $\deg^+(w) \geq k + 1$, contradicting k -regularity. Hence $N^{++}(v) = V \setminus (\{v\} \cup N^+(v))$, which has size $n - 1 - k = k$ since $n = 2k + 1$. $\square$

Neither item is load-bearing for Theorem 1: the only tournament-forced shard in range, $n15_s7$, finished UNSAT and its DRAT proof is drat-trim-verified, so at $n = 15$ Fisher’s theorem and Lemma T are independent confirmations, not dependencies. They are load-bearing only for the $n = 17$ tournament shard $n17_s8$ discussed in Section 5.2, whose SAT run hit the solver cap with no verdict; there the closure comes from the published literature [2], with Lemma T as the unreviewed in-house alternative. We keep Lemma T documented because it is short enough to audit in one sitting, while noting it is now redundant for the headline claim.

3 Two independent pipelines

3.1 Pipeline A: SAT with DRAT certificates

Every instance in Table 1 was solved by CaDiCaL 3.0.1 [13] with binary DRAT proof output. Every completed proof was checked with drat-trim [14] built from source (commit 2e3b2dc, upstream marijnheule/drat-trim). All 59 completed refutations have **s VERIFIED** receipts, including the two giants: *n15_s7* (0.60 GB proof, verified in 1,689 s) and *n16_s7* (4.23 GB proof, verified in 9,671 s). No receipt has ever failed. The receipts were *regenerated from scratch in the write-up session* (2026-07-23) by a different agent than the one that ran the solver, and are stored alongside this note (**receipts/**). As a negative control, a truncated copy of one proof was rejected by drat-trim (**s NOT VERIFIED**), confirming the checker does not rubber-stamp.

Additionally, all 63 CNF files on disk were regenerated byte-identically from **encode.py** with their documented flags in the write-up session, tying the artifacts that were solved to the encoder that was audited: the four whole-space files ($n = 6, \dots, 9$) with no flags, the three cross-check files with **--mindeg** and full-lex flags, and all 56 shard files with **--mindeg d --exactdeg0 d --pin-nbhd --sb-mode twoblock**.

3.2 Pipeline B: exhaustive enumeration, $n \leq 8$

Independently of SAT, encoder, and solver: nauty’s [18] **geng** generated all graphs, **directg -o** all their orientations, and a from-scratch C checker (**check_d6.c**, about 80 lines, reading digraph6 directly and re-checking the no-loop/no-digon property itself) tested every oriented graph on $n \leq 8$ vertices. Counts match OEIS A001174 [19] (number of oriented graphs on n unlabeled vertices) exactly at every order, which is strong evidence the enumeration is complete:

n	1	2	3	4	5	6	7	8
oriented graphs	1	2	7	42	582	21,480	2,142,288	575,016,219
counterexamples	0	0	0	0	0	0	0	0

The $n \leq 7$ runs were re-executed live during the write-up session (seconds); the $n = 8$ run was sharded ten ways (**geng’s res/mod** splitting) and the per-shard **checked** counts sum to exactly 575,016,219. A positive control (**check_weak**, the same checker with the comparison deliberately weakened) flags all 7 oriented graphs on 3 vertices, as it must.

So for $n \leq 8$ the result is double-proved by disjoint toolchains (PySAT/CaDiCaL/drat-trim versus nauty/hand-written C), and for $n = 9, \dots, 16$ it rests on Pipeline A plus the written arguments of Section 2.

3.3 Auxiliary lane: Cayley digraphs of small groups

As a targeted probe outside the SAT chain (not part of the headline claim), all 55 groups of orders 12 through 24 were constructed and brute-force verified to be groups, with isomorphism-type counts per order matching the classification (5, 1, 2, 1, 14, 1, 5, 1, 5, 2, 2, 1, 15); for each group, every connection set S with $S \cap S^{-1} = \emptyset$ was enumerated (1,505,160 sets total) and the resulting Cayley digraph checked at the identity vertex, vertex-transitivity justifying the single-vertex check; that shortcut was itself revalidated by full all-vertex checks on every 250th set. Zero counterexamples. This lane shares no code with either pipeline above but has not been independently re-audited; treat it as a search result, not a certified one.

4 Controls

Beyond the negative proof-checker control and the byte-identical CNF regeneration already described:

- **Slack controls (SAT expected).** Relaxing the per-vertex bound by one (encoding $|N^{++}(v)| \leq |N^+(v)|$) must be satisfiable, since, e.g., directed cycles and their blowups sit exactly on the boundary. This was confirmed at $n = 3$ (plain) and, critically, at $n = 6$ and $n = 12$ *through the full shard machinery* (pinning, exact degree, two-block lex): the shard path is not accidentally over-constrained. The $n = 12$ control model was decoded back into an explicit digraph and checked to be a genuine boundary witness, validating the totalizer calibration end to end.
- **Shard UNSAT control.** A shard instance at $n = 7$, where Pipeline B independently proves no counterexample exists, is UNSAT as expected. All four control instances were re-solved from the stored CNF files during the write-up session with the expected verdicts (SAT, SAT, UNSAT, SAT).
- **Solver agreement.** Spot-checked shards were re-solved with Glucose 4.2 and MiniSat 2.2 with agreeing verdicts (no proofs logged from those; CaDiCaL’s DRAT proofs are the receipts).

5 Results

Table 1 lists every SAT instance with its DIMACS size, CaDiCaL wall time, DRAT proof size, verdict, and this session’s drat-trim receipt. The table is generated mechanically from the artifacts by `gen_tables.py`; no number in it was typed by hand. Wall times are from a heavily shared machine and should be read as upper bounds. Shard counts per order equal $\lfloor (n-1)/2 \rfloor$ exactly: 4 shards at $n = 10$, then 5, 5, 6, 6, 7, 7, 8, 8 through $n = 18$.

Table 1: All SAT instances. Columns: DIMACS size, CaDiCaL wall time (seconds), DRAT proof size, solver verdict, and the drat-trim receipt regenerated in the write-up session (2026-07-23).

instance	meaning	vars	clauses	time	proof	verdict	drat-trim
n6	$n=6$ whole space, $\delta^+ \geq 1$, no SB	264	621	<0.1	–	UNSAT	VERIFIED
n7	$n=7$ whole space, $\delta^+ \geq 1$, no SB	392	1015	0.4	–	UNSAT	VERIFIED
n8	$n=8$ whole space, $\delta^+ \geq 1$, no SB	544	1540	10	–	UNSAT	VERIFIED
n9	$n=9$ whole space, $\delta^+ \geq 1$, no SB	720	2214	279	–	UNSAT	VERIFIED
n10_d1_sb	$n=10$ whole space, $\delta^+ \geq 1$, full lex	1246	4902	27	–	UNSAT	VERIFIED
n10_d3_sb	$n=10$, $\delta^+ \geq 3$ (lemma), full lex	1536	5552	15	–	UNSAT	VERIFIED
n11_d3_sb	$n=11$, $\delta^+ \geq 3$ (lemma), full lex	1942	7264	503	–	UNSAT	VERIFIED
n10_s1	$n=10$, shard $d_0=1$	1207	4571	<0.1	–	UNSAT	VERIFIED
n10_s2	$n=10$, shard $d_0=2$	1497	5221	<0.1	–	UNSAT	VERIFIED
n10_s3	$n=10$, shard $d_0=3$	1497	5221	<0.1	–	UNSAT	VERIFIED
n10_s4	$n=10$, shard $d_0=4$	1497	5221	<0.1	–	UNSAT	VERIFIED
n11_s1	$n=11$, shard $d_0=1$	1526	6031	<0.1	–	UNSAT	VERIFIED
n11_s2	$n=11$, shard $d_0=2$	1900	6900	<0.1	–	UNSAT	VERIFIED
n11_s3	$n=11$, shard $d_0=3$	1900	6900	<0.1	–	UNSAT	VERIFIED
n11_s4	$n=11$, shard $d_0=4$	1900	6900	<0.1	–	UNSAT	VERIFIED
n11_s5	$n=11$, shard $d_0=5$	1900	6900	<0.1	–	UNSAT	VERIFIED

instance	meaning	vars	clauses	time	proof	verdict	drat-trim
n12_s1	$n=12$, shard $d_0=1$	1881	7747	<0.1	–	UNSAT	VERIFIED
n12_s2	$n=12$, shard $d_0=2$	2349	8875	<0.1	–	UNSAT	VERIFIED
n12_s3	$n=12$, shard $d_0=3$	2349	8875	<0.1	–	UNSAT	VERIFIED
n12_s4	$n=12$, shard $d_0=4$	2349	8875	<0.1	–	UNSAT	VERIFIED
n12_s5	$n=12$, shard $d_0=5$	2349	8875	0.2	–	UNSAT	VERIFIED
n13_s1	$n=13$, shard $d_0=1$	2272	9737	<0.1	–	UNSAT	VERIFIED
n13_s2	$n=13$, shard $d_0=2$	2844	11167	<0.1	–	UNSAT	VERIFIED
n13_s3	$n=13$, shard $d_0=3$	2844	11167	<0.1	–	UNSAT	VERIFIED
n13_s4	$n=13$, shard $d_0=4$	2844	11167	<0.1	–	UNSAT	VERIFIED
n13_s5	$n=13$, shard $d_0=5$	2844	11167	0.7	–	UNSAT	VERIFIED
n13_s6	$n=13$, shard $d_0=6$	2844	11167	10.0	–	UNSAT	VERIFIED
n14_s1	$n=14$, shard $d_0=1$	2699	12019	<0.1	–	UNSAT	VERIFIED
n14_s2	$n=14$, shard $d_0=2$	3385	13797	<0.1	–	UNSAT	VERIFIED
n14_s3	$n=14$, shard $d_0=3$	3385	13797	<0.1	–	UNSAT	VERIFIED
n14_s4	$n=14$, shard $d_0=4$	3385	13797	<0.1	–	UNSAT	VERIFIED
n14_s5	$n=14$, shard $d_0=5$	3385	13797	2.3	–	UNSAT	VERIFIED
n14_s6	$n=14$, shard $d_0=6$	3385	13797	41	–	UNSAT	VERIFIED
n15_s1	$n=15$, shard $d_0=1$	3162	14611	<0.1	–	UNSAT	VERIFIED
n15_s2	$n=15$, shard $d_0=2$	3972	16786	<0.1	–	UNSAT	VERIFIED
n15_s3	$n=15$, shard $d_0=3$	3972	16786	<0.1	–	UNSAT	VERIFIED
n15_s4	$n=15$, shard $d_0=4$	3972	16786	<0.1	–	UNSAT	VERIFIED
n15_s5	$n=15$, shard $d_0=5$	3972	16786	2.8	–	UNSAT	VERIFIED
n15_s6	$n=15$, shard $d_0=6$	3972	16786	51	–	UNSAT	VERIFIED
n15_s7	$n=15$, shard $d_0=7$	3972	16786	4643	–	UNSAT	VERIFIED
n16_s1	$n=16$, shard $d_0=1$	3661	17531	<0.1	–	UNSAT	VERIFIED
n16_s2	$n=16$, shard $d_0=2$	4605	20155	<0.1	–	UNSAT	VERIFIED
n16_s3	$n=16$, shard $d_0=3$	4605	20155	<0.1	–	UNSAT	VERIFIED
n16_s4	$n=16$, shard $d_0=4$	4605	20155	<0.1	–	UNSAT	VERIFIED
n16_s5	$n=16$, shard $d_0=5$	4605	20155	3.9	–	UNSAT	VERIFIED
n16_s6	$n=16$, shard $d_0=6$	4605	20155	116	–	UNSAT	VERIFIED
n16_s7	$n=16$, shard $d_0=7$	4605	20155	19080	–	UNSAT	VERIFIED
n17_s1	$n=17$, shard $d_0=1$	4196	20797	<0.1	–	UNSAT	VERIFIED
n17_s2	$n=17$, shard $d_0=2$	5284	23925	<0.1	–	UNSAT	VERIFIED
n17_s3	$n=17$, shard $d_0=3$	5284	23925	<0.1	–	UNSAT	VERIFIED
n17_s4	$n=17$, shard $d_0=4$	5284	23925	<0.1	–	UNSAT	VERIFIED
n17_s5	$n=17$, shard $d_0=5$	5284	23925	2.6	–	UNSAT	VERIFIED
n17_s6	$n=17$, shard $d_0=6$	5284	23925	84	–	UNSAT	VERIFIED
n17_s7	$n=17$, shard $d_0=7$	5284	23925	–	–	unknown	–
n17_s8	$n=17$, shard $d_0=8$	5284	23925	–	–	unknown	–
n18_s1	$n=18$, shard $d_0=1$	4804	24464	<0.1	–	UNSAT	VERIFIED
n18_s2	$n=18$, shard $d_0=2$	6064	28172	<0.1	–	UNSAT	VERIFIED
n18_s3	$n=18$, shard $d_0=3$	6064	28172	<0.1	–	UNSAT	VERIFIED
n18_s4	$n=18$, shard $d_0=4$	6064	28172	<0.1	–	UNSAT	VERIFIED
n18_s5	$n=18$, shard $d_0=5$	6064	28172	2.9	–	UNSAT	VERIFIED
n18_s6	$n=18$, shard $d_0=6$	6064	28172	113	–	UNSAT	VERIFIED
n18_s7	$n=18$, shard $d_0=7$	6064	28172	–	–	unknown	–
n18_s8	$n=18$, shard $d_0=8$	6064	28172	–	–	unknown	–

5.1 Fully verified claim

For $n \leq 16$: every listed instance is UNSAT with a drat-trim-verified DRAT proof. By the completeness argument of Section 2.1 (for $n \leq 9$; the $n = 10$ whole-space instance additionally uses the full-lex soundness argument of Section 2.2) and the covering argument of Section 2.2 (for the sharded orders), this proves Theorem 1. For $n \leq 8$, Pipeline B independently confirms

it.

5.2 Per-order status

- **$n = 15$:** all 7 shards UNSAT by CaDiCaL (top shard s_7 : 4,643s wall, 0.60 GB proof), all 7 DRAT proofs drat-trim-verified (s_7 receipt: **s VERIFIED**, 1,689 s). Shard s_7 forces a 7-regular tournament, so $n = 15$ is closed twice over: by the verified DRAT proof and, independently, by Fisher’s theorem [2] (or the unreviewed Lemma 1). Part of the headline claim.
- **$n = 16$:** all 7 shards UNSAT by CaDiCaL (s_7 : 19,080s wall, 4.23 GB proof), all 7 DRAT proofs drat-trim-verified (s_7 receipt: **s VERIFIED**, 9,671 s). No tournament argument applies at even n ; the certificate chain alone carries $n = 16$, and it is now fully proof-verified. Part of the headline claim.
- **$n = 17, 18$:** shards s_1 through s_6 UNSAT and verified. The near-tournament shards ($n17_s7$, $n17_s8$, $n18_s7$, $n18_s8$) were attempted and hit the solver cap with verdict **c UNKNOWN**: as machine results they are open. $n17_s8$ forces an 8-regular tournament and is closed by Fisher [2] (or Lemma 1) regardless of its SAT run; that closure comes from the literature, not from a certificate. So $n = 17$ is open only at shard s_7 , while $n = 18$ is open at shards s_7 and s_8 ; neither order is claimed.

6 What an external reviewer must audit

This note was produced by an automated pipeline. Before any submission, a human must independently check, at minimum:

1. **The soundness chain (Section 2).** The DRAT certificates prove only that specific CNF files are unsatisfiable. The encoding-completeness argument, the sink and arc-count bounds on d_0 , and the two-block lex-leader covering argument are pen-and-paper claims living outside all certificates. They are short; audit every line.
2. **encode.py against Section 2.1.** About 150 lines. Confirm the clause groups match the stated encoding and that PySAT’s totalizer (`CardEnc.atmost`, `EncType.totalizer`) is used correctly (auxiliary variables must not be shared across constraints incorrectly; the slack controls test this but do not prove it).
3. **Lemma 1**, or simply replace every use of it by the citation to Fisher [2] and delete it.
4. **Re-verification of all certificates on independent hardware**, rebuilding CaDiCaL and drat-trim from upstream sources. The full artifact set is about 28 GB; SHA-256 checksums of every CNF, every completed proof, and the three load-bearing source files are in `receipts/SHA256SUMS.txt`. Ideally re-check with a second, independently implemented DRAT checker (e.g., the formally verified `cake_lpr`).
5. **Pipeline B.** Read `check_d6.c` (about 80 lines, including its own digraph6 parser and digon check); confirm the `geng/directg -o` invocations enumerate exactly the oriented graphs; re-derive the OEIS comparison.

6. **Provenance discipline.** Only `encode.py`, `check_d6.c`, the CNF/DRAT artifacts, and the receipts are load-bearing. Earlier search code from the same overnight runs (local search, circulant search, the round-1 SAT stack) is untrusted by policy and cited nowhere in the argument. Confirm nothing in the chain silently depends on it.
7. **Literature claims.** Re-check the history of Section 1 against the primary sources, in particular the exact statement of Kaneko–Locke and the status of the 2026 minimum out-degree 7 preprint, and re-run the search for any published exhaustive small-order verification we may have missed.
8. **The $n = 15, 16$ upgrade.** The two large proof verifications completed with `s VERIFIED` receipts (`n15_s7`: 1,689s; `n16_s7`: 9,671s), and the headline of this note was moved to $n \leq 16$ on that basis, by the automated pipeline. A human should re-read those two receipts (`receipts/n15_s7.drattrim.txt`, `receipts/n16_s7.drattrim.txt`) and confirm the wording of the upgraded claim.

Reproduction appendix

All commands assume macOS with Homebrew; everything is CPU-only. On a shared machine, prefix heavy runs with `nice -n 15`.

1. Toolchain:

```
brew install cadical nauty python
pip install python-sat
git clone https://github.com/marijnheule/drat-trim && make -C drat-trim
```

2. Whole-space instances (assumption-free), $n = 6, \dots, 9$:

```
python3 encode.py N --out nN.cnf
cadical -q nN.cnf nN.drat # expect: s UNSATISFIABLE
drat-trim/drat-trim nN.cnf nN.drat # expect: s VERIFIED
```

3. Shard instances, $n = 10, \dots, 18$, $d = 1, \dots, \lfloor (n-1)/2 \rfloor$:

```
python3 encode.py N --mindeg D --exactdeg0 D --pin-nbhd \
  --sb-mode twoblock --out nN_sD.cnf
cadical -q nN_sD.cnf nN_sD.drat
drat-trim/drat-trim nN_sD.cnf nN_sD.drat
```

4. Whole-space cross-checks at $n = 10, 11$: `--mindeg 1 --sb-mode full` (and `--mindeg 3` for the lemma-conditioned variants).
5. Controls: `--slack` variants must come back SAT; truncating any `.drat` must make `drat-trim` report `s NOT VERIFIED`.
6. Pipeline B ($n \leq 8$; shard $n = 8$ via `res/mod`):

```
cc -O2 -o check_d6 check_d6.c
geng -q N | directg -o -q | ./check_d6
geng -q 8 r/10 | directg -o -q | ./check_d6 # r = 0..9, sum counts
```

7. Regenerate Table 1 from the artifacts, and re-verify every stored proof:

```
python3 gen_tables.py
sh verify_all_drat.sh
```

References

- [1] N. Dean and B. J. Latka, Squaring the tournament: an open problem, *Congressus Numerantium* **109** (1995), 73–80.
- [2] D. C. Fisher, Squaring a tournament: a proof of Dean’s conjecture, *Journal of Graph Theory* **23** (1996), 43–48.
- [3] F. Havet and S. Thomassé, Median orders of tournaments: a tool for the second neighborhood problem and Sumner’s conjecture, *Journal of Graph Theory* **35** (2000), 244–256.
- [4] Y. Kaneko and S. C. Locke, The minimum degree approach for Paul Seymour’s distance 2 conjecture, *Congressus Numerantium* **148** (2001), 201–206.
- [5] G. Chen, J. Shen, and R. Yuster, Second neighborhood via first neighborhood in digraphs, *Annals of Combinatorics* **7** (2003), 15–20.
- [6] D. Fidler and R. Yuster, Remarks on the second neighborhood problem, *Journal of Graph Theory* **55** (2007), 208–220.
- [7] J. Brantner, G. Brockman, B. Kay, and E. Snively, Contributions to Seymour’s second neighborhood conjecture, *Involve* **2** (2009), 387–395.
- [8] Z. Cohn, A. Godbole, E. Wright Harkness, and Y. Zhang, The number of Seymour vertices in random tournaments and digraphs, *Graphs and Combinatorics* **32** (2016), 1805–1816.
- [9] A. Espuny Díaz, A. Girão, B. Granet, and G. Kronenberg, Seymour’s second neighbourhood conjecture: random graphs and reductions, *Random Structures & Algorithms* (2025), arXiv:2403.02842.
- [10] A. Sadhukhan, R. B. Sandeep, and S. Sen, A proof of Seymour’s second neighborhood conjecture for oriented graphs with minimum out-degree equal to 7, arXiv:2606.30588 (2026). Preprint, unrefereed as far as we know.
- [11] B. Mohar, Seymour’s Second Neighborhood Conjecture, problem page, https://www.sfu.ca/~mohar/Problems/P0601_SeymourSecondNbdConj.html, accessed 2026-07-23.
- [12] D. B. West, Seymour’s 2nd Neighborhood Conjecture, open problem page, <https://dwest.web.illinois.edu/openp/2ndnbhd.html>, accessed 2026-07-23.
- [13] A. Biere, K. Fazekas, M. Fleury, and M. Heisinger, CaDiCaL, Kissat, Paracooba, Plingeling and Treengeling entering the SAT Competition 2020, in *Proc. SAT Competition 2020*, University of Helsinki, 2020. Version used here: CaDiCaL 3.0.1.

- [14] N. Wetzler, M. J. H. Heule, and W. A. Hunt Jr., DRAT-trim: efficient checking and trimming using expressive clausal proofs, in *Theory and Applications of Satisfiability Testing (SAT 2014)*, LNCS 8561, Springer, 2014, 422–429.
- [15] A. Ignatiev, A. Morgado, and J. Marques-Silva, PySAT: a Python toolkit for prototyping with SAT oracles, in *SAT 2018*, LNCS 10929, Springer, 2018, 428–437.
- [16] O. Bailleux and Y. Boufkhad, Efficient CNF encoding of Boolean cardinality constraints, in *Principles and Practice of Constraint Programming (CP 2003)*, LNCS 2833, Springer, 2003, 108–122.
- [17] J. Crawford, M. Ginsberg, E. Luks, and A. Roy, Symmetry-breaking predicates for search problems, in *Proc. 5th International Conference on Principles of Knowledge Representation and Reasoning (KR'96)*, 1996, 148–159.
- [18] B. D. McKay and A. Piperno, Practical graph isomorphism, II, *Journal of Symbolic Computation* **60** (2014), 94–112.
- [19] OEIS Foundation Inc., entry A001174 (number of oriented graphs on n nodes), *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org/A001174>.